

DACsys Cryogenic Data Acquisition & Control System

Modular High-Resolution Logging for Cryogenic & Quantum Experiments

Precision Data Acquisition.
Modular Cryogenic Control.
Built for Quantum Research.



DACsys is a modular data acquisition and control platform optimized for continuous, high-resolution data logging in cryogenic and quantum experiments.

It enables stable, low-noise monitoring of multiple experimental parameters—including voltage, current, resistance, and temperature—over extended measurement durations. The system is designed for seamless integration with laboratory instrumentation, supporting real-time analysis and control in complex experimental environments.

KEY FEATURES

- **Multi-Channel Data Acquisition**
Simultaneously acquires voltage, current, resistance, and temperature data for multi-parameter experimental correlation with scalable channel expansion.
- **Continuous Real-Time Logging**
Enables uninterrupted long-duration data capture with optimized stability and consistent dataset integrity over extended measurement cycles.
- **Low-Noise Analog Front-End**
Provides precision analog signal measurement with preserved signal fidelity for low-current and low-voltage regimes while minimizing noise and drift.
- **Modular & Scalable Architecture**
Features an expandable and configurable system design adaptable to increasing experimental complexity and evolving research requirements.
- **Real-Time Data Integration**
Supports direct data streaming to LabVIEW, Python, and MATLAB for live analysis, visualization, feedback control, and automated workflows.

Notes

- System configuration depends on application-specific requirements
- Optimal performance achieved with proper shielding and grounding practices

APPLICATIONS

- Cryogenic experiment monitoring
- Quantum system data acquisition
- Multi-parameter experimental analysis
- Long-duration measurement logging
- Device characterization and testing
- Automated laboratory measurement systems



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SPECIFICATIONS

DATA ACQUISITION CAPABILITIES

Measurement Types	Voltage, current, resistance, temperature
Channel Architecture	Multi-channel, scalable
Acquisition Mode	Simultaneous multi-parameter logging
Resolution	High-resolution (configuration dependent)

PERFORMANCE & SIGNAL INTEGRITY

Analog Front-End	Low-noise precision design
Signal Fidelity	Optimized for low-level measurements
Measurement Stability	High stability over long durations
Noise Performance	Low-noise operation

SAMPLING & LOGGING

Sampling Mode	Continuous real-time acquisition
Sampling Rate	Configurable
Logging Duration	Long-duration / continuous
Data Integrity	Stable uninterrupted logging

SYSTEM ARCHITECTURE

Design Type	Modular, scalable
Expansion	Channel and functionality expansion supported
Integration	Compatible with multi-instrument setups
Configuration	Customizable system architecture

INTERFACE & CONTROL

Data Interface	USB / system-dependent
Software Support	LabVIEW, Python, MATLAB
Data Streaming	Real-time export and monitoring
Automation	Full integration into automated workflows

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